

Proposed Tendler-Style Replacement for Extension Across the Boundary

Standalone preview — not incorporated into Chapter 3

July 2026

This document is a preview of the proposed replacement only. It makes no change to the existing text of Chapter 3.

Proposed introductory description

Second, on the geometric generic Abel–Jacobi fiber, the boundary is the hyperplane at infinity. A Tendler-style prime-to- p power argument, together with the structure of the abelian fundamental group of affine space, forces the corresponding boundary inertia representation to be trivial. Purity then extends $\mathcal{F}^{[d]}$ uniquely across the boundary.

Proposed replacement for the extension argument

Lemma 1. *The rank-one local system $\mathcal{F}^{[d]}$ on $U^{(d)}$, which is tame along the boundary H , extends uniquely to a rank-one local system on $\tilde{C}^{(d)}$.*

Proof. Let ξ be the generic point of $\mathrm{Pic}_{\tilde{C}}^d$, let $\Omega = k(\xi)^{\mathrm{sep}}$, and consider the geometric generic fiber of the Abel–Jacobi morphism. By the projective-bundle description,

$$\tilde{C}_{\Omega}^{(d)} \simeq \mathbb{P}_{\Omega}^r, \quad H_{\Omega} \simeq \mathbb{P}_{\Omega}^{r-1},$$

and hence

$$U_{\Omega}^{(d)} = \tilde{C}_{\Omega}^{(d)} \setminus H_{\Omega} \simeq \mathbb{A}_{\Omega}^r.$$

Let

$$\chi: \pi_1(\mathbb{A}_{\Omega}^r) \longrightarrow \Lambda^{\times}$$

be the character corresponding to the restriction of $\mathcal{F}^{[d]}$ to this fiber.

If $\mathrm{char}(k) = 0$, then

$$\pi_1(\mathbb{A}_{\Omega}^r) = 1,$$

so χ is trivial [1, Example 4.14].

Suppose now that $\mathrm{char}(k) = p > 0$. Let I_{∞} denote the inertia group at the generic point of the hyperplane at infinity. Since $\mathcal{F}^{[d]}$ is tame along H , the finite group $\chi(I_{\infty})$ has order prime to p . Let N be its exponent. Then $(N, p) = 1$, and the character χ^N is trivial on I_{∞} .

Consequently, the local system corresponding to χ^N is unramified at the generic point of $\mathbb{P}_{\Omega}^{r-1}$. By purity, it extends to $\mathbb{P}_{\Omega}^r$. Since

$$\pi_1(\mathbb{P}_{\Omega}^r) = 1,$$

this extension is constant, and therefore

$$\chi^N = 1.$$

On the other hand, χ factors through

$$\pi_1(\mathbb{A}_\Omega^r)^{\text{ab}},$$

which is a pro- p group [1, Example 4.15]. Hence the finite group $\text{im}(\chi)$ is a p -group. Since its exponent divides N , and N is prime to p , it follows that

$$\text{im}(\chi) = 1.$$

Thus χ is trivial.

Let η be the generic point of H . The preceding argument shows that the pullback of the finite inertia representation at η to the geometric generic fiber is trivial. The corresponding extension of henselian discrete valuation rings has ramification index 1 and separable residue-field extension. Triviality of a finite inertia representation is detected after such a base change. Hence the inertia group at η acts trivially on $\mathcal{F}^{[d]}$.

Therefore $\mathcal{F}^{[d]}$ is unramified at the generic point of H . Since $\tilde{C}^{(d)}$ is regular and H is the only codimension-one component of $\tilde{C}^{(d)} \setminus U^{(d)}$, purity gives an extension of $\mathcal{F}^{[d]}$ to $\tilde{C}^{(d)}$. The extension is unique because

$$\pi_1(U^{(d)}) \longrightarrow \pi_1(\tilde{C}^{(d)})$$

is surjective. □

Remark. The argument does not assert that tame ramification is generally unramified. Tameness is used only to choose N prime to p . The conclusion that the inertia character is trivial then follows from the global geometry of the geometric generic fiber and from the fact that the abelian fundamental group of affine space is pro- p .

References

- [1] Avichai Tendler, *Geometric Class Field Theory*, Master's thesis, Hebrew University of Jerusalem, 2010; arXiv:1507.00104 (2015), especially Examples 4.14–4.15 and §9.3. <https://arxiv.org/abs/1507.00104>.